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General Introduction

In today's fast-paced digital world, patients and healthcare professionals increasingly seek accessible, efficient, and secure ways to deliver care beyond the walls of hospitals and clinics. Teleconsultation has emerged as a practical response to this need, enabling medical follow-up, first opinions, and continuity of care while reducing unnecessary travel and waiting time. Yet, many existing solutions remain fragmented, complex for non-technical users, or insufficiently aligned with clinical workflows and privacy requirements. To address these challenges, developing a dedicated teleconsultation module offers a streamlined and user-friendly path to remote medical care.

Our teleconsultation module provides a centralized space where patients can find the right practitioner, request an appointment, and join a secure video session with minimal effort. From the first click to the end of the consultation, the experience is designed to be simple and reassuring: clear scheduling, intuitive guidance to connect, and secure exchange of medical documents when needed.

Beyond the consultation itself, the module supports the essential steps that make remote care clinically useful: pre-consultation information gathering, documentation during the session, and structured follow-up. It also emphasizes security and confidentiality through authentication, role-based access, encrypted communications, and traceability mechanisms that protect sensitive health information.

The platform is built around three main actors. Patients benefit from quick access to care, appointment reminders, and a smooth connection experience. Doctors manage their availability, conduct consultations, and produce clinical notes and recommendations. Administrators supervise user accounts, permissions, configuration, and audit trails to ensure the service remains reliable and compliant.

The goal of this project is to design and develop a comprehensive, intuitive teleconsultation module that improves access to care without compromising medical quality or privacy. By prioritizing usability, security, and workflow alignment, the proposed solution aims to make remote consultations a natural extension of everyday healthcare services.

This report consists of four chapters. Chapter 1 presents the project context, motivates the choice of teleconsultation, and summarizes the adopted methodology and planning. Chapter 2 defines the stakeholders, functional and non-functional requirements, and illustrates the expected behavior through modeling and mockups. Chapter 3 focuses on the conceptual design of the solution. Finally, Chapter 4 covers the implementation phase, including the development environment, system architecture, and key interfaces, concluding with perspectives for future improvements.

Part I

...

Chapter 1

Project Presentation

1.1 Introduction

In recent years, the healthcare industry has undergone a massive digital transformation, driven by the increasing demand for flexible and accessible medical solutions. The rise of teleconsultation has highlighted the importance of efficient remote care systems that bridge the gap between patients and practitioners. This project aims to develop a modern teleconsultation module that addresses current technical gaps by providing a human-centered and technically robust solution for virtual medical care.

1.2 Presentation of the host organization

1.2.1 Mediwave

1.2.2 Services of Mediwave

1.3 Project Presentation

1.3.1 Problematic

Despite the rapid expansion of digital health platforms, the transition from physical to virtual medical consultations is still hindered by significant technical and structural flaws. Patients and doctors frequently face synchronization issues where calls fail to connect despite both parties being online, often resulting in unfair "no-show" charges. Furthermore, the administrative burden on doctors is significant; the absence of automated tools for recording and summarizing consultations forces practitioners to spend excessive time on paperwork. Therefore, there is a clear need for a secure, synchronized, and AI-enhanced teleconsultation module that optimizes the experience for all users.

1.3.2 Existing Work Analysis

The development of digital technologies has significantly transformed the healthcare sector, particularly through the emergence of teleconsultation platforms. These platforms allow patients to consult healthcare professionals remotely, facilitating access to medical services and improving the flexibility of healthcare delivery.

In Tunisia, telemedicine initiatives have started to appear in recent years, offering new possibilities for remote medical interaction between patients and healthcare providers. Some platforms provide basic online medical advice or limited consultation services, contributing to the gradual adoption of digital healthcare solutions.

At the international level, several teleconsultation platforms have reached a more advanced stage of development by integrating various functionalities such as video consultations, digital prescriptions, appointment scheduling, and patient medical records.

Analysis of Existing Work

To develop a high-performing teleconsultation module, we evaluated three major platforms currently leading the market:

- **Doqtoor.com**: A Tunisian telemedicine platform founded in 2020, offering two core services: a free asynchronous Q&A forum where patients get answers from volunteer physicians, and paid video coaching sessions with healthcare professionals. It targets Arabic and French-speaking patients in underserved areas across North Africa.

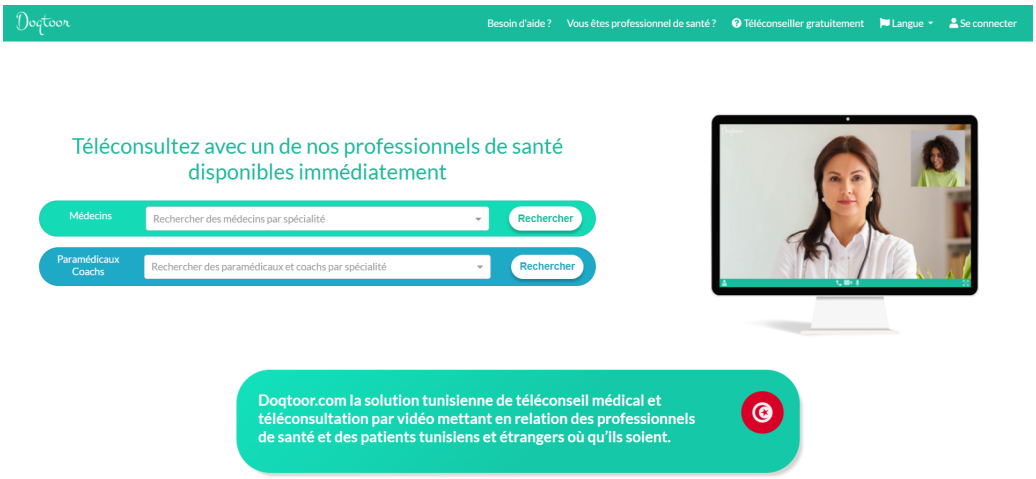


Figure 1.1: Doqtoor.com website

- **Qare.fr**: A French teleconsultation platform founded in 2017, connecting patients with 2,500+ doctors across 50+ specialties via on-demand or scheduled video consultations. It issues digital prescriptions and sick leave certificates, maintains a patient dossier, and consultations are reimbursable by French health insurance.



Figure 1.2: Qare.fr website

- **MedecinDirect.fr**: A French telemedicine platform offering video, phone, and written consultations available 24/7. It uses a team-based care model where nurses, GPs, and specialists collaborate on the same case. Over 16 million patients access it free through partner health insurance companies.



Figure 1.3: MedecinDirect.fr website

Critique of Existing Work

- **Doqtoor.com (Tunisia)**
  - **Strengths**
    - **First Tunisian telemedicine platform** — A pioneer in the Tunisian digital health market with no local competitors at launch.
    - **Community-driven volunteer doctor network** — Volunteer physicians keep operational costs low while maintaining broad patient coverage.
    - **Regional accessibility for Arabic-speaking users** — One of the few platforms built specifically for Arabic/French-speaking North African patients.
    - **Dual model: free Q&A + paid video coaching** — Serves patients across all income levels through a free advice tier and affordable paid consultations.

  - **Limitations**
    - **No mobile app** — Web-only access creates a poor experience for smartphone users, who represent the majority of Tunisian internet users.
    - **No digital prescriptions** — The absence of a licensed physician teleconsultation module prevents the platform from issuing legally valid medical prescriptions.
    - **No patient record** — Medical history and previous consultations are not stored, preventing continuity of care between consultations.
    - **Limited doctors and appointments** — The platform lacks a specialist directory, appointment scheduling system, and doctor selection feature.

- **Qare.fr (France)**
  - **Strengths**
    - **Large network of healthcare professionals** — The platform connects patients with more than 2,500 doctors across over 50 medical specialties, reducing waiting times.
    - **Full insurance reimbursement** — Consultations are covered by the French Assurance Maladie and most mutuelle insurance providers, removing financial barriers.
    - **Rich feature set** — The platform provides digital prescriptions, sick leave certificates, and an integrated patient medical record.
  - **Limitations**
    - **Hidden service fees** — Some consultations involve additional non-reimbursable fees that are not clearly visible before booking.
    - **No AI features** — The platform lacks AI-based functionalities such as automatic consultation summarization to assist practitioners in reviewing and documenting patient interactions.
    - **Limited accessibility for international users** — The platform is designed mainly for the French healthcare system and does not support non-French-speaking users.

- **MédecinDirect.fr (France)**
  - **Strengths**
    - **True 24/7 availability** — The platform offers medical consultations at any time, including nights, weekends, and public holidays.
    - **Multiple consultation channels** — Patients can consult healthcare professionals through video calls, phone calls, or written messaging.
    - **Partnerships with insurance companies** — More than 16 million insured patients can access consultations at no direct cost through partner mutuelles.
  - **Limitations**
    - **Patient can't choose their doctor** — Doctors are automatically assigned, preventing patients from building long-term relationships with a specific practitioner.

    - **Complex onboarding** — The activation code system for mutuelle users and the unclear process for uninsured patients create friction during registration.
    - **No AI features** — The platform does not provide AI-based tools such as automatic consultation summarization to support practitioners.
    - **Slow written consultation responses** — Asynchronous messages may take several hours, which is not suitable for patients seeking quick medical advice.

1.3.3 Proposed Solution

To overcome these limitations, we propose a specialized teleconsultation module integrated into our web application. The solution focuses on five main pillars:

- **Mobile-first application** — Build native iOS and Android apps from day one, prioritising the mobile experience.
- **Digital prescription module** — Allow certified physicians to issue legally valid electronic prescriptions directly through the platform.
- **Centralised patient record** — Store consultation history, prescriptions, and documents in a persistent and secure patient dossier.
- **Specialist directory with scheduling** — Provide searchable doctor profiles with specialty filters, ratings, and real-time appointment booking.
- **Full pricing transparency** — Display a clear cost breakdown before booking, distinguishing reimbursable charges from out-of-pocket fees.
- **AI-powered consultation summarization** — Automatically generate a concise summary of each consultation, highlighting key symptoms, recommendations, and prescribed treatments.
- **Doctor selection and continuity of care** — Allow patients to choose their practitioner and easily rebook the same doctor for follow-up consultations.
- **Response time guarantee** — Ensure a maximum response time of two hours for written consultations, with automatic escalation if no response is provided.

1.4 Work Methodology

Before undertaking an IT project, it is necessary to choose a working methodology to ensure development aligns with client requirements. We have chosen the Agile Scrum methodology because its characteristics match our desired approach, and it is adopted by all teams in the company.

1.4.1 Agile Methodology

The Agile methodology is an iterative and incremental approach to software development, characterized by its flexibility and ability to adapt to changes. Instead of following a strict plan, Agile teams encourage continuous collaboration between team members and stakeholders, along with short development cycles called "iterations" or "sprints". These iterations allow for functional versions of the software to be created at regular intervals, enabling quick feedback and its integration into the development process. The Agile approach emphasizes customer satisfaction, rapid delivery of high-quality software, and the ability to effectively respond to changing needs and priorities. t is based on values such as collaboration, adaptability, transparency, and self-organizing teamwork.

1.4.2 Agile Methodology: SCRUM

Scrum (see Figure 1.4) is an Agile project management method that enables rapid product delivery and modification through its various roles, tools, and a set of meetings that result in the delivery of small, functional software pieces to the client at regular intervals. It accommodates changes that occur during development without causing significant issues.

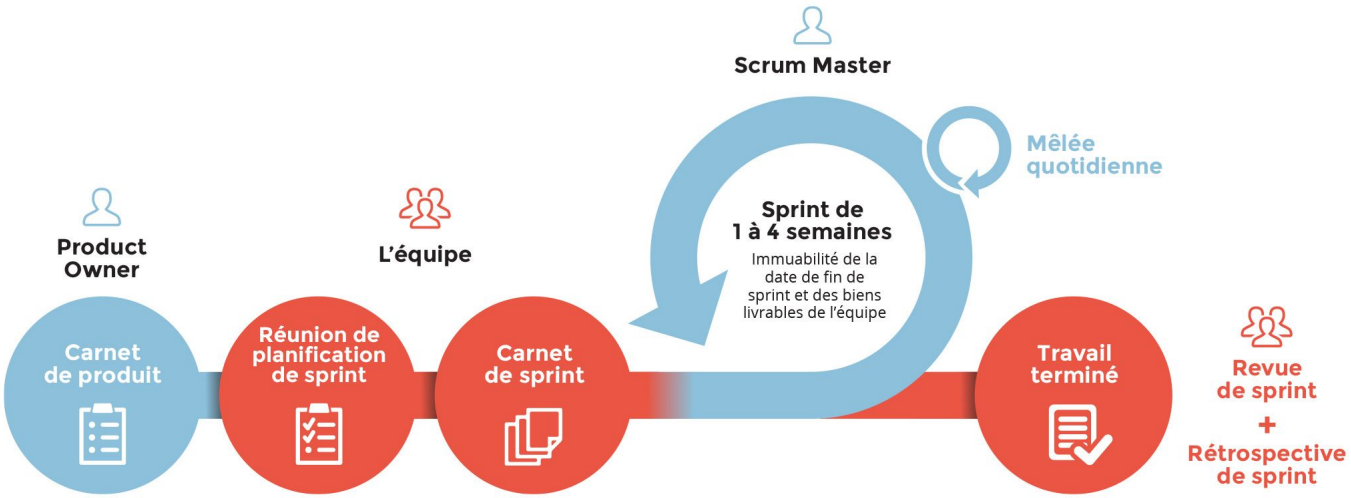


Figure 1.4: Lifecycle in SCRUM

1.4.3 Advantages of the SCRUM Model

Among the advantages of the SCRUM model, we highlight:

- **Flexibility and adaptability:** SCRUM allows constant flexibility and adaptability, meeting changing requirements and ensuring the product aligns with stakeholder needs.

- **Improved collaboration and communication:** SCRUM fosters close collaboration, regular communication, and transparency between team members and stakeholders, promoting shared understanding, better teamwork, and quick problem resolution.

- **Faster time to market:** SCRUM enables rapid and iterative delivery of usable product increments, providing quick feedback, incremental updates, and continuous customer satisfaction.

1.4.4 The SCRUM Team

The SCRUM team is divided into three main roles that participate in the product development process:

- **The Product Owner:** The Product Owner represents the client's interests, defines project requirements, sets priorities, validates increments, and actively participates in planning and review meetings, ensuring alignment with the client's business expectations and objectives.

- **The SCRUM Master:** The SCRUM Master ensures proper implementation of the SCRUM methodology, facilitates problem-solving, acts as a coach, and promotes communication and collaboration within the team to achieve project goals.

- **The SCRUM Development Team:** The SCRUM Development Team is a self-organizing, self-managing, multidisciplinary team responsible for creating and implementing product features. It collaborates closely, sharing knowledge and responsibilities to achieve sprint goals and improve coordination and efficiency.

1.5 Work planning

1.6 Conclusion

In this chapter, we introduced the context and challenges that motivate the development of our teleconsultation module, examined the strengths and limitations of existing telemedicine solutions, and defined a secure and user-centered approach to remote care. We also presented the Agile Scrum methodology adopted for the project and summarized our work planning through the Gantt chart. The next chapter moves from vision to construction, focusing on the design choices that will turn these requirements into a reliable, clinician-friendly, and patient-ready solution.

Chapter 2

Analyse et spécification des besoins

2.1 Introduction

In this chapter, we translate the teleconsultation vision into clear, actionable requirements. We identify who interacts with the module, what each actor expects to accomplish, and the quality constraints that guarantee a safe and smooth medical experience. Finally, we use modeling artifacts to make the module's behavior explicit and to validate our understanding of the main interactions and processes before moving to design and implementation.

2.2 Requirements Specification

2.2.1 Actor Identification

The key actors in our teleconsultation system are:

- **Administrator:** Manages platform operations, verifies medical credentials, and oversees the general technical stability of the system.
- **Visitor:** An unauthenticated user who can browse the medical directory, search for specialists by category, and create an account.
- **Patient:** Inherits all actions from the Visitor. Additionally, they can manage their medical profile, book appointments, and participate in secure teleconsultations.
- **Doctor (Practitioner):** Conducts remote consultations, manages availability, and utilizes AI-assisted tools for clinical documentation.

2.2.2 Functional Requirements

Functional requirements represent the specific actions the system must perform to meet the needs of each actor.

1. **Administrator:** As the platform manager, the administrator must ensure high service quality and security. Their functional requirements include:

    - User Management: The administrator must be able to create, verify, and moderate both patient and doctor accounts.
    - Directory Oversight: Ability to manage medical specialties and ensure practitioners are correctly listed in the search engine.
    - System Performance Monitoring: Ability to track consultation logs and ensure the real-time synchronization systems are operational.

2. **Patient:** To ensure a high-quality user experience and autonomy, the patient's functional requirements include:

    - Doctor Selection: Access to a comprehensive directory to browse professional profiles and **choose a specific doctor**, addressing the personalization gap found in existing solutions.
    - Secure Consultation: Ability to join video calls and private chats through a secure, synchronized module.
    - Booking Management: Ability to schedule, view, or cancel consultations with real-time availability updates.
    - Profile Management: Users can update personal health information and view their consultation history.

3. **Doctor:** As the service provider, the doctor needs tools that reduce administrative load and ensure technical reliability. Their functional requirements include:

    - AI-Powered Summarization: Access to a tool that automatically generates clinical summaries from recorded consultations to assist in follow-ups.
    - Real-time Heartbeat Monitoring: A system (powered by Socket.io) that verifies the patient is active in the waiting room, eliminating technical "no-show" errors.
    - Consultation Dashboard: A native WebRTC interface to conduct stable, browser-based video calls without needing third-party software.
    - Availability Management: Ability to set consultation hours and manage their patient queue efficiently.

2.2.3 Non-Functional Requirements

Non-functional requirements describe the quality constraints and technical standards that ensure the system's reliability:

- **Confidentiality and Security:** The platform must guarantee the absolute privacy of medical data. All video streams and chats must be encrypted, and data sovereignty must be maintained via native integration.
- **Synchronization Reliability:** The website must deliver real-time accuracy. Using a "heartbeat" mechanism ensures that both parties are synchronized, preventing technical glitches and unfair billing.
- **Interface Ergonomics:** The module should provide an intuitive and accessible interface for both elderly patients and busy practitioners, ensuring that the focus remains on healthcare rather than technology.
- **Performance and Scalability:** The system must provide low-latency video and audio, even during peak hours, and be capable of scaling as more doctors join the medical network.

2.3 Requirements Modeling

2.3.1 UML Modeling Language

UML (Unified Modeling Language) (see Figure 2.1) is a standard graphical modeling language used in the software development industry to visually represent computer systems. It allows describing different system components such as use cases, classes, sequences, activities, components, etc. Using UML enables better system understanding and facilitates communication between different project stakeholders. It is therefore a valuable tool for the design, analysis, and documentation of computer systems.



Figure 2.1: UML Logo

2.3.2 General Use Case Diagrams

In this section, we present the general use case diagram that groups all basic use cases to provide an overview of our platform's operation:

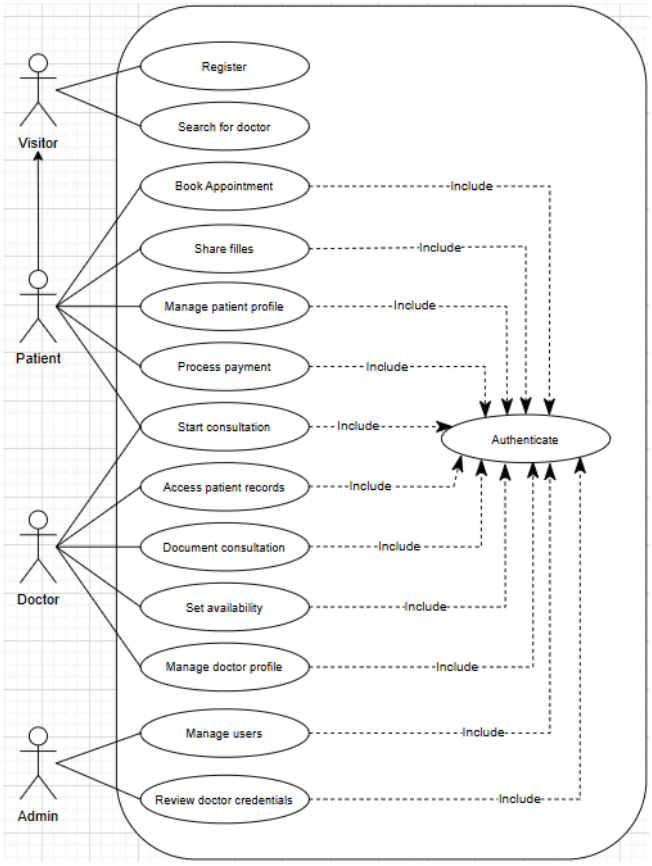


Figure 2.2: Global Use Case Diagram

2.3.3 Use Case Refinement

To better understand the operation of a use case, we will present descriptive tables for the main use cases.

Use Case Analysis: «Authenticate»

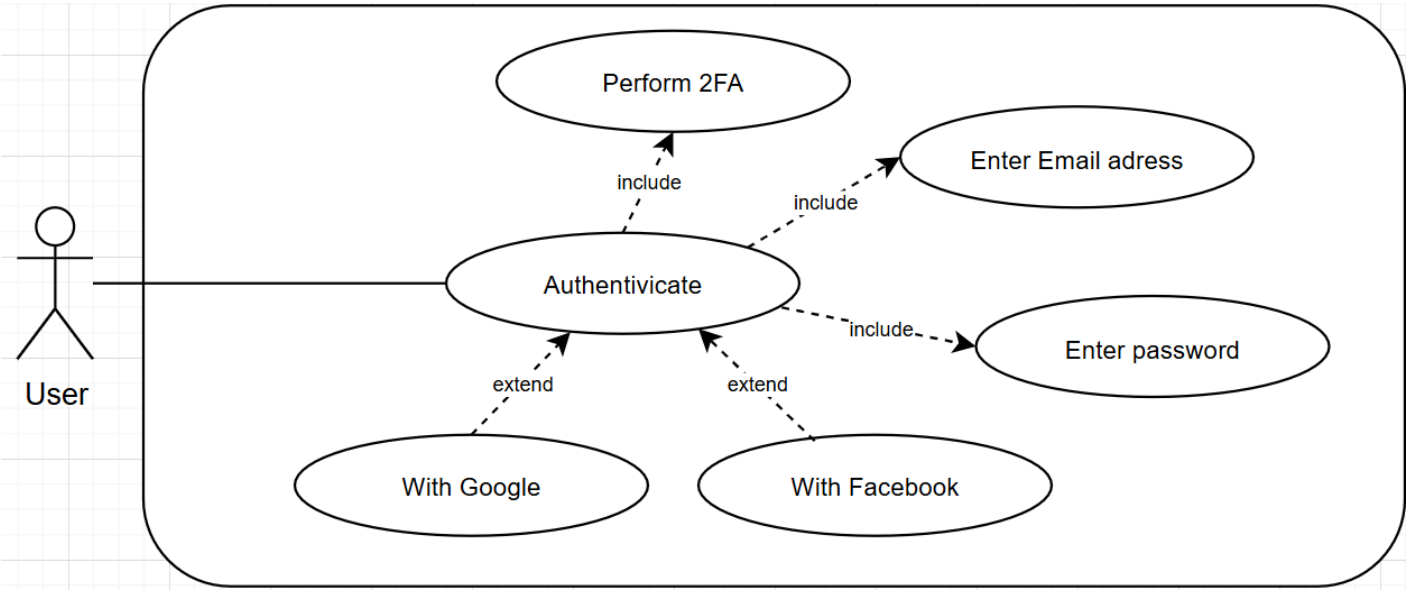


Figure 2.3: Use Case Diagram «Authenticate»

Table 2.1 shows the textual description of the «Authentication» use case:

Use Case Name	Authentication
Diagram Reference	Figure 2.3
Primary Actor	User
Secondary Actors	Google Authentication Service, Facebook Authentication Service, SMS/Email OTP Service
Preconditions	• User has an existing account (for email/password login) • User has a valid Google or Facebook account (for social login) • System is available and connected to the internet
Basic Flow	1. User selects "Authenticate" 2. System prompts user to enter email address («include») 3. User enters email address 4. System prompts user to enter password («include») 5. User enters password 6. System validates credentials 7. System prompts for Two-Factor Authentication («include») 8. User enters the verification code sent to their device 9. System validates the 2FA code 10. System grants access to the user
Alternative Flows	• **Login with Google («extend»)**: User selects "With Google"; system authenticates via Google service and then proceeds to step 7 (2FA). • **Login with Facebook («extend»)**: User selects "With Facebook"; system authenticates via Facebook service and then proceeds to step 7 (2FA). • **Invalid Credentials**: System displays error message and requests re-entry of email/password. • **Invalid 2FA Code**: System displays error message and allows the user to resend or re-enter the code.
Postconditions	• User is successfully authenticated and identity is verified • User session is created with high-security clearance • User is redirected to the dashboard/home page

Tableau 2.1: Use Case Description: Authentication with 2FA

Use Case Analysis: «Booking management system»

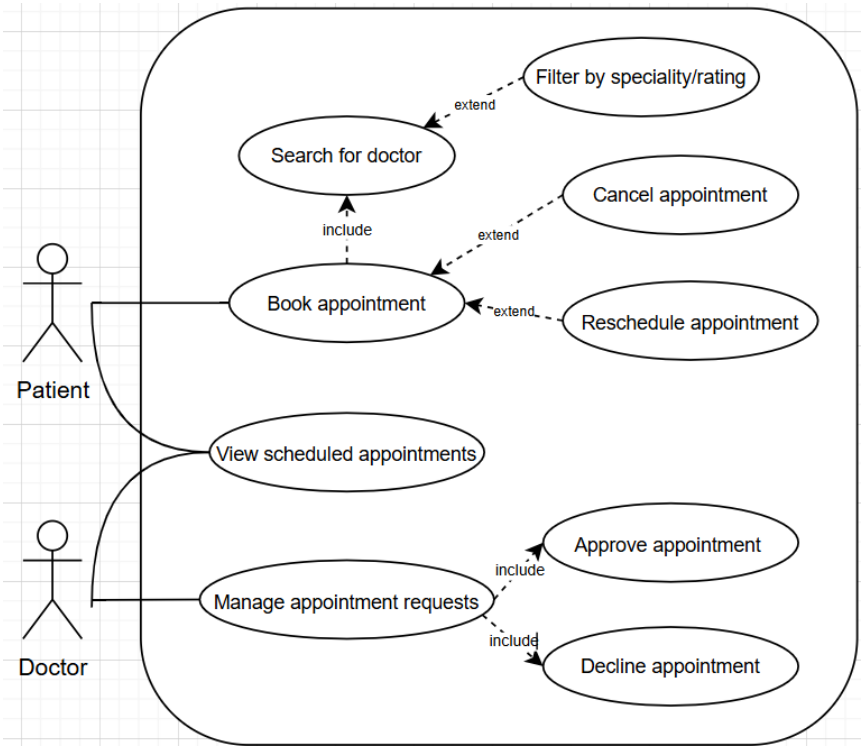


Figure 2.4: Use Case Diagram «Booking management system»

Table 2.2 shows the textual description of the «Booking management system» use case:

Use Case Name	Appointment Booking System
Diagram Reference	Figure 2.4
Primary Actor	Patient
Secondary Actors	Doctor
Preconditions	• Patient and Doctor are authenticated and logged in • Doctor profiles exist in the system • System is available and operational
Basic Flow	1. Patient selects "Book Appointment" 2. System triggers "Search for Doctor" («include») 3. Patient selects a doctor and confirms the booking 4. Doctor receives the request and selects "Manage Appointment Requests" 5. System triggers "Approve Appointment" («include») 6. Patient is notified of the approved appointment 7. Patient can view the appointment via "View Scheduled Appointments"
Alternative Flows	• **Filter by Speciality/Rating («extend»)**: Patient filters doctors by speciality or rating during search. • **Cancel Appointment («extend»)**: Patient cancels an existing appointment. • **Reschedule Appointment («extend»)**: Patient moves the appointment to a different date/time. • **Decline Appointment («include»)**: Doctor rejects the request; patient is notified of the decline. • **No Available Slots**: System notifies the patient and suggests alternatives.
Postconditions	• Appointment status is updated in the system • Both Patient and Doctor are notified of the final appointment status • Patient can view all scheduled appointments

Tableau 2.2: Use Case Description: Appointment Booking System

Use Case Analysis: «Start consultation»

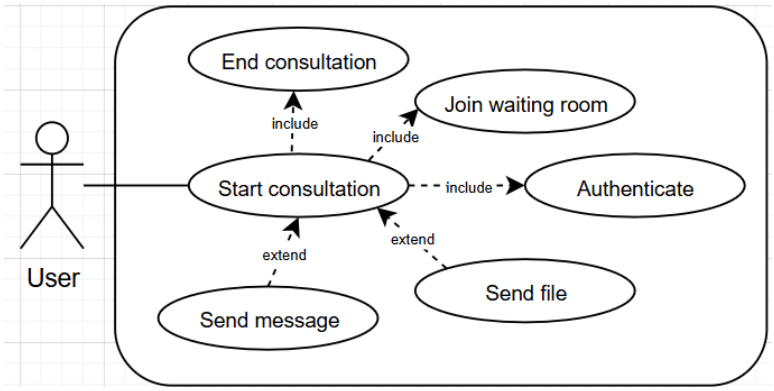


Figure 2.5: Use Case Diagram «Start consultation»

Table 2.3 shows the textual description of the «Start consultation» use case:

Use Case Name	Start consultation
Diagram Reference	Figure 2.5
Primary Actor	User
Secondary Actors	Authentication Service, Waiting Room Management System
Preconditions	• User has access to the telehealth platform. • An appointment is scheduled or the queue is active.
Basic Flow	1. User selects the option to "Start consultation". 2. System triggers "Authenticate" (<<include>>) to verify user identity. 3. System triggers "Join waiting room" (<<include>>) to enter the session queue. 4. System establishes the live connection between participants. 5. System triggers "End consultation" (<<include>>) when the session is concluded.
Alternative Flows	• **Send message (<<extend>>)**: User sends text-based communications during the active session. • **Send file (<<extend>>)**: User uploads or shares documents/images during the consultation. • **Authentication Failure**: Access is denied if the included "Authenticate" step fails.
Postconditions	• Consultation session is logged in the system history. • Any shared messages or files are saved to the session record. • User is moved out of the active waiting room.

Tableau 2.3: Use Case Description: Start Consultation

Use Case Analysis: «Document consultation»

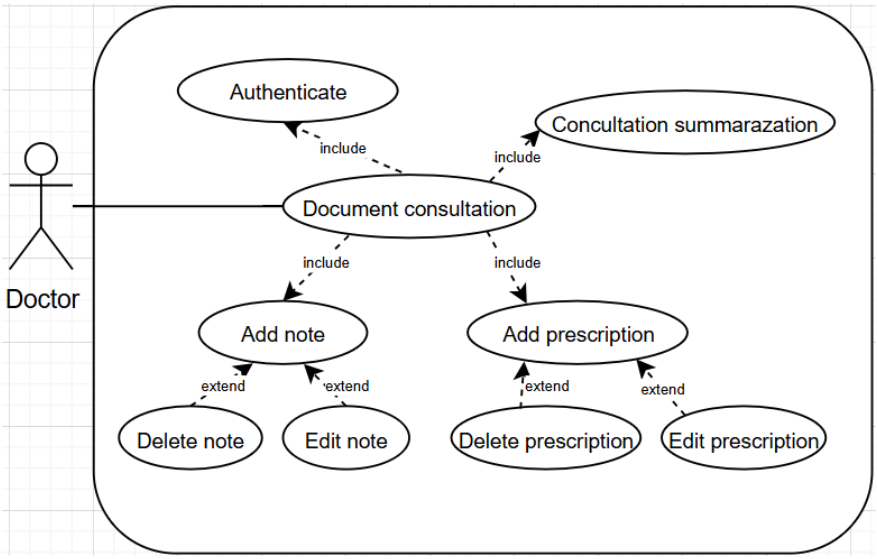


Figure 2.6: Use Case Diagram «Document consultation»

Table 2.4 shows the textual description of the «Document consultation» use case:

Use Case Name	Document consultation
Diagram Reference	Figure 2.6
Primary Actor	Doctor
Secondary Actors	Patient Records Database (Implicit)
Preconditions	• Doctor is present and has access to the system. • A consultation session has been initiated or completed.
Basic Flow	1. Doctor selects the option to "Document consultation". 2. System triggers "Authenticate" (<<include>>) to verify credentials. 3. System triggers "Add note" (<<include>>) for clinical observations. 4. System triggers "Add prescription" (<<include>>) for medication management. 5. System triggers "Consultation summarization" (<<include>>) to finalize the record. 6. System saves the completed documentation to the patient's history.
Alternative Flows	• **Edit/Delete Note (<<extend>>)**: Doctor modifies or removes existing notes during the documentation process. • **Edit/Delete Prescription (<<extend>>)**: Doctor modifies or removes medication entries before final submission. • **Authentication Failure**: System denies access if credentials provided in the "Authenticate" step are invalid.
Postconditions	• Consultation details are securely stored. • Patient record is updated with new notes and prescriptions. • A summary of the visit is generated.

Tableau 2.4: Use Case Description: Document Consultation

Use Case Analysis: «Manage users»

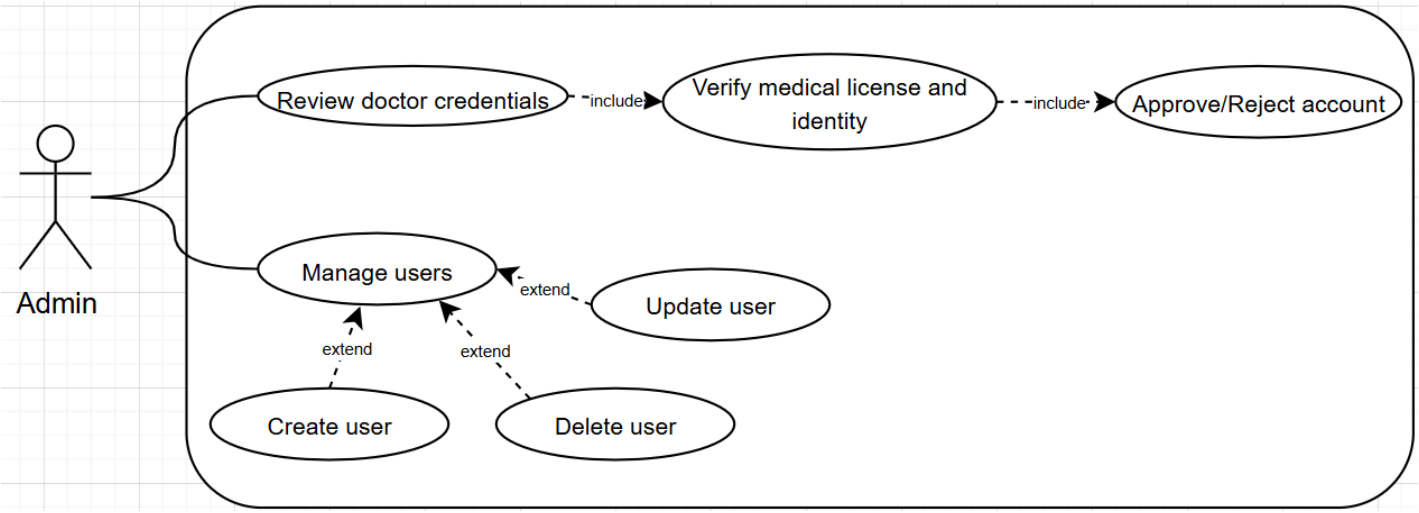


Figure 2.7: Use Case Diagram «Document consultation»

Table 2.5 shows the textual description of the «Manage users» use case:

Use Case Name	Review doctor credentials / Manage users
Diagram Reference	Figure 2.7
Primary Actor	Admin
Secondary Actors	Licensing Board Database, User Management System
Preconditions	• Admin is logged into the administrative portal. • New doctor accounts are pending review or existing user accounts require modification.
Basic Flow	1. Admin selects "Review doctor credentials". 2. System triggers "Verify medical license and identity" (<<include>>). 3. System triggers "Approve/Reject account" (<<include>>) based on verification results. 4. Admin accesses "Manage users" to view the general user list.
Alternative Flows	• **Create user** (<<**extend**>>): Admin manually adds a new user to the system. • **Update user** (<<**extend**>>): Admin modifies details of an existing user account. • **Delete user** (<<**extend**>>): Admin removes a user account from the system. • **Verification Failed**: If the medical license cannot be verified, the account is automatically flagged for rejection.
Postconditions	• Doctor accounts are activated or deactivated based on credential status. • User database is updated with created, modified, or deleted entries.

Tableau 2.5: Use Case Description: Administrative Management

2.4 Mock-up scenario

2.5 Conclusion

Part II

...

Chapter 3

Étude conceptuelle

3.1 Introduction

After analyzing the needs, we begin in this chapter the design phase. We focus on designing an appropriate structure for the teleconsultation module . This step is crucial to the smooth progress of the project and aims to detail the tasks to be undertaken and to prepare the ground for the implementation phase.

3.2 Conception de la plateforme : vue dynamique

3.2.1 Diagramme de séquences

3.2.2 Diagramme PLUS

3.3 Conception de la plateforme : vue statique

3.3.1 Diagramme de classes

3.3.2 Diagramme PLUS

3.4 Conclusion

Chapter 4

Réalisation

4.1 Introduction

4.2 Environnement de développement

4.2.1 Environnement technique

4.2.2 Environnement technique

4.2.3 Environnement matériel

4.3 Présentation des interfaces du plateforme

4.4 Conclusion

Conclusion générale

Conclusion

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